

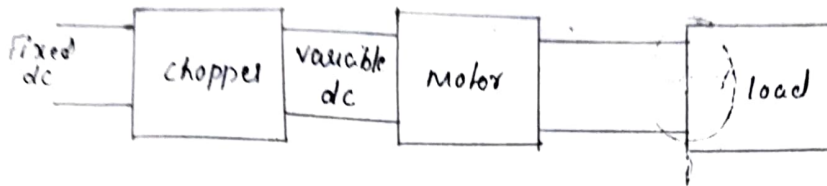


KTU NOTES

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Raisa Mol K.K
S7 666-

DC motor speed can be controlled through DC chopper. Fig shows simple block diagram of DC chopper drive system. Fixed DC voltage is fed to the DC chopper circuit. DC chopper converts fixed DC into variable DC voltage. This variable DC voltage is fed to the motor. By varying the DC voltage, the motor speed can be controlled.

Advantages of DC chopper control

1. High efficiency
2. Flexibility in control
3. Light weight
4. Small size
5. Quick response
6. Regeneration down to very low speeds

The chopper offers a number of advantages over controlled rectifiers for a DC motor control in open loop and closed loop configurations. Because of the higher frequency of the o/p voltage ripple, the ripple in the motor armature current is less, and the region of discontinuous conduction in the speed-torque plane is smaller. A reduction in armature current ripples reduces the armature losses. A reduction of discontinuous conduction region improves speed regulation and transient response.

Application of DC

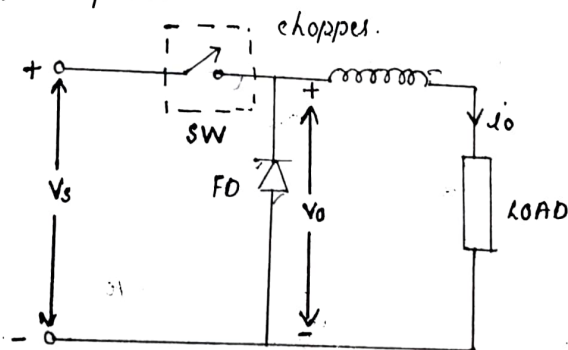
1. Industrial drive
2. Battery operated vehicles
3. Traction motor control in electrical traction
4. Trolley cars
5. Hoists
6. Electric braking

CHOPPERS

A chopper is a static device that converts fixed dc input voltage to variable dc output voltage directly. Chopper may be thought of as dc equivalent of an ac transformer since they behave in an identical manner. The power semiconductor devices used for a chopper circuit can be power BJT, power MOSFET, GTO or force-commutated thyristor.

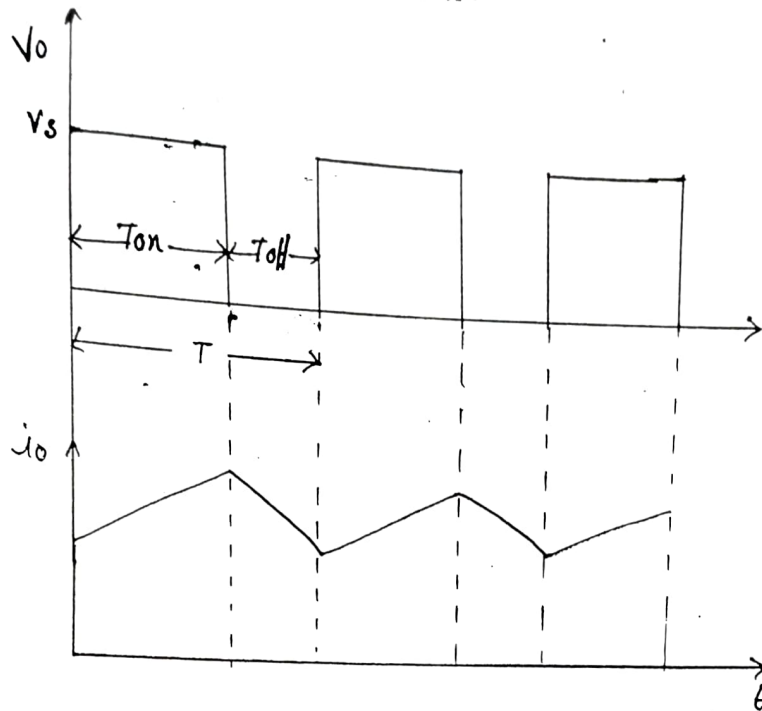
Principle of chopper operation

A chopper is thyristor on/off switch that connects the load to source and disconnects the load from source at a fast speed.



The chopper is represented by a switch SW. During the period T_{on} , the chopper is ON and the supply terminals are connected to the load terminals. During the period T_{off} the chopper is OFF, the load current flows through the freewheeling diode, and the load terminals are

Therefore chopped dc voltage is induced at the load terminals.



The average load voltage.

$$V_o = \frac{1}{T} \int_0^T V_s dt$$

$$V_o = \frac{1}{T} \int_0^{T_{on}} V_s dt$$

$$V_o = \frac{T_{on} V_s}{T}$$

$$V_o = \frac{1}{T} \int_0^{T_{on}} V_s dt$$

$$= \frac{1}{T} [V_s]_0^{T_{on}}$$

$$V_o = \frac{T_{on} V_s}{T} = \frac{T_{on} V_s}{T_{on} + T_{off}}$$

$T \rightarrow$ Total time period

$T_{on} \rightarrow$ Total on time period

$T_{off} \rightarrow$ Total off time period

$$T = \frac{1}{f}$$

$f \rightarrow$ chopping frequency.

$$V_o = \alpha V_s = \delta V_s$$

$$\delta = \alpha = \frac{T_{on}}{T}$$

is the duty cycle

The load voltage can be controlled by varying the duty cycle α (s)

The average output voltage can be controlled by varying α by opening and closing the semiconductor switch periodically. The various control strategies for varying duty cycle α or δ are as follows

- 1) Time Ratio Control method (TRC)
- 2) Current limit Control method (CLC).

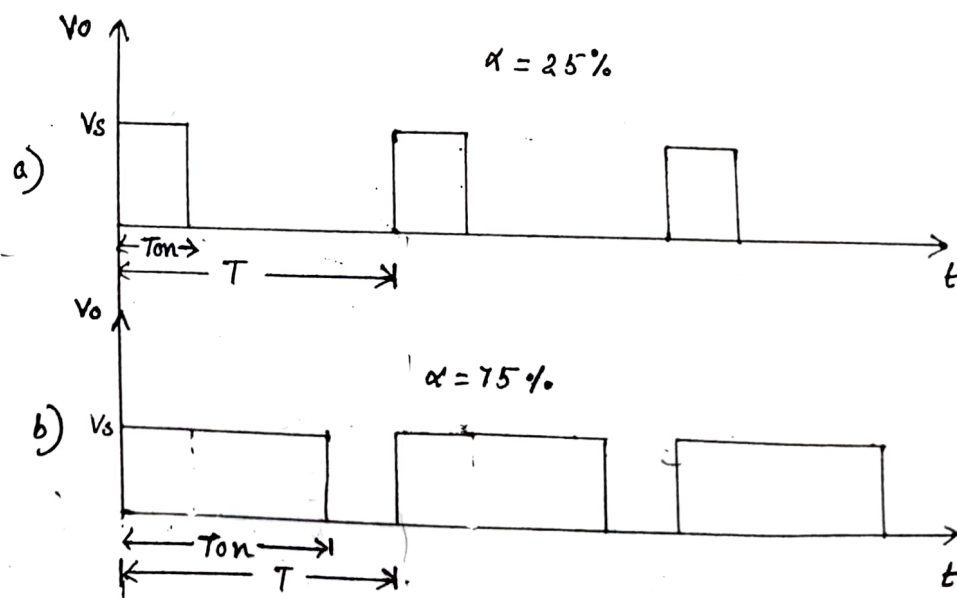
I Time Ratio Control method.

In the Time Ratio Control, the value of $\frac{T_{on}}{T}$ is varied. This is effected in two ways

- a) Constant frequency TRC control
- b) Variable frequency TRC control

i) constant frequency control. (Pulse width modulation control)

In this control method ON time T_{on} is varied but chopping frequency (chopping period) is kept constant. The width of the pulse is varied and this type of control is known as Pulse width modulation (PWM).



shows illustrates the principle of pulse width modulation. Here the chopping period T is constant. In fig (a) $T_{on} = \frac{1}{4}T$ so that $\alpha = 25\%$. In fig (b) $T_{on} = \frac{3}{4}T$, so that $\alpha = 75\%$. α can be varied from zero to unity.

Therefore output voltage V_o can be varied between zero to source voltage V_s . The constant frequency control gives low ripple and requires smaller size of filter and fast response. Thus this scheme is preferred for chopper drives.

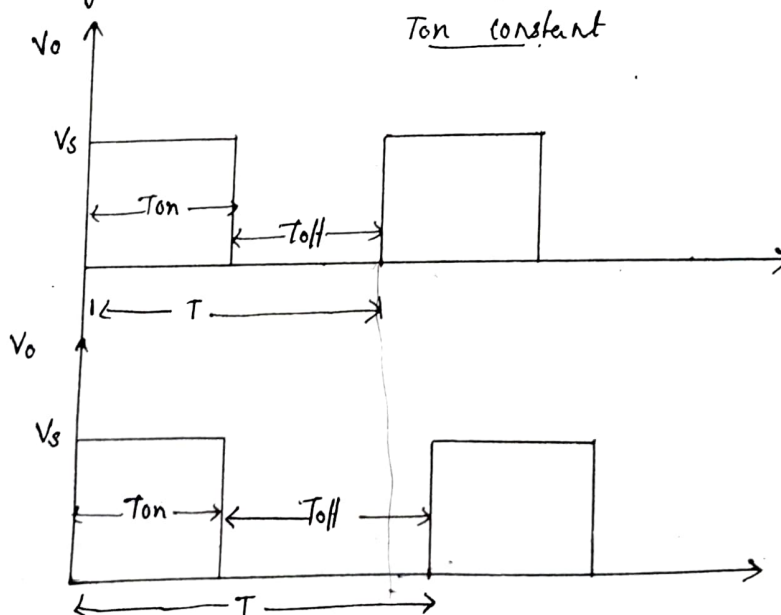
2) Variable Frequency systems.

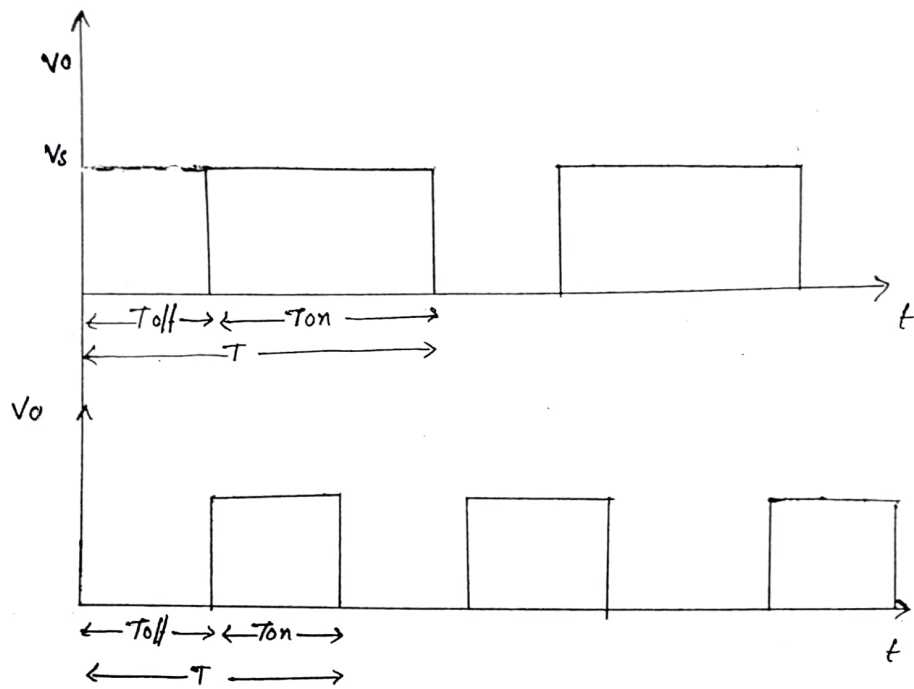
In this control method, the chopping frequency f (or chopping period T) is varied and either.

- 1) On-time T_{on} is kept constant
- 2) Off-time T_{off} is kept constant

This method of controlling duty cycle α (%) is called frequency modulation scheme.

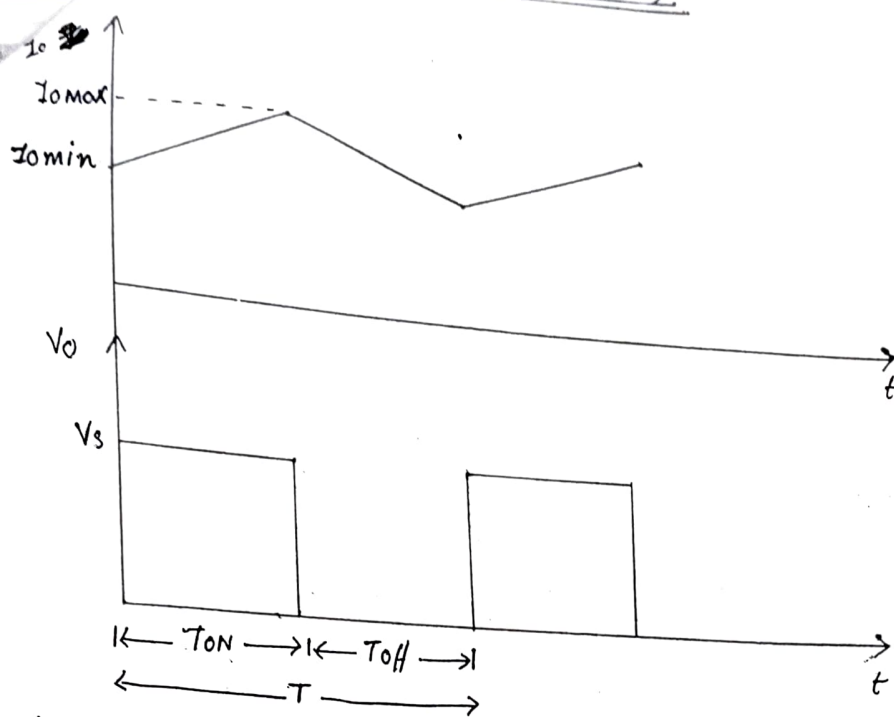
Fig (1) T_{on} is kept constant but T is varied





Frequency modulation control method has some disadvantages as compared to pulse width modulation.

1. The chopping frequency has to be varied over a wide range for the control ~~voltage~~ of output voltage in frequency modulation. The filter design is very difficult for wide range of frequency variation.
2. This type of control would generate harmonics at unpredictable frequencies which would produce interference with signalling and telephone lines.
3. The large off-time in FM scheme may make the load current discontinuous which is undesirable.



In this strategy, the chopper is switched ON and OFF so that the current in the load is maintained between two limits (I_{omin} to I_{omax}). The chopper is switched ON when the output current equal to I_{omin} (lower limit). The chopper is kept closed till load current increases upper limit (I_{omax}). When the current exceeds upper limit, the chopper is switched OFF. During OFF period, the load current free wheels and decreases exponentially. When the load current reaches the lower limit, the chopper is switched ON.

current limit control method is possible for constant frequency and variable frequency systems. It is used only when load has storage elements.

According to operation choppers are classified as

1. 1st quadrant type chopper (motoring operation)
2. 2nd quadrant type chopper (Braking operation)
3. ~~four~~ ^{four} quadrant type chopper (motoring & braking operation)
- 4) Two quadrant type chopper.

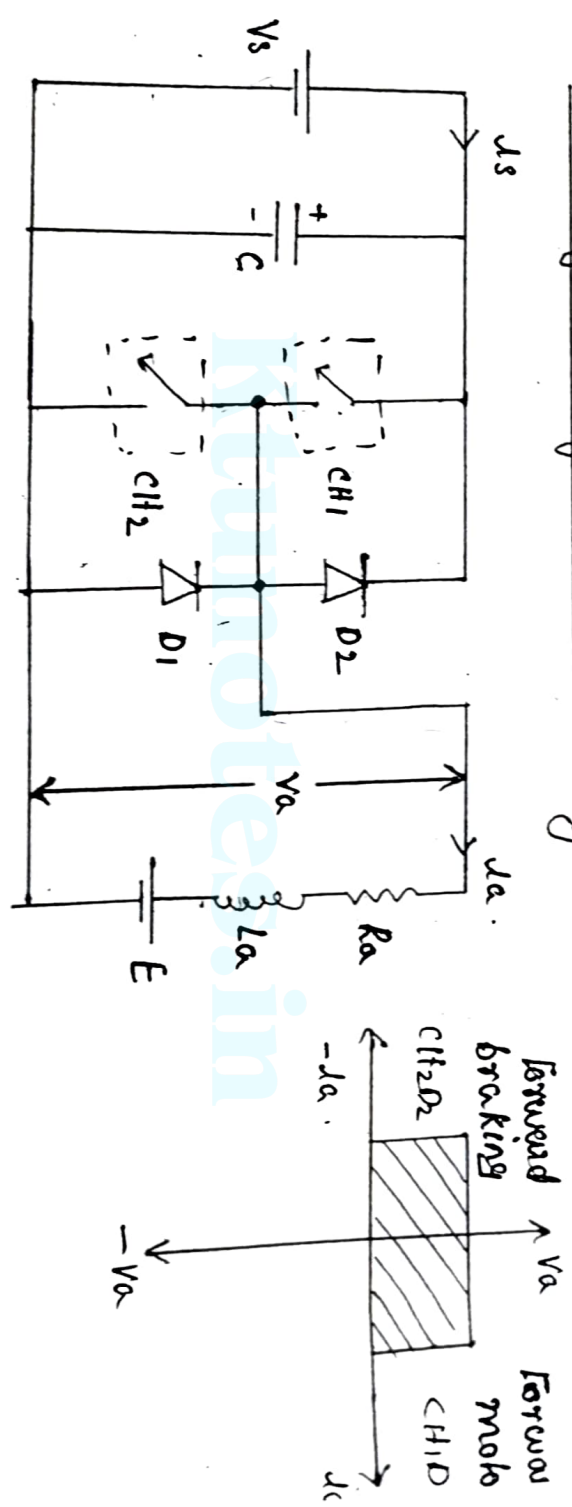
-T

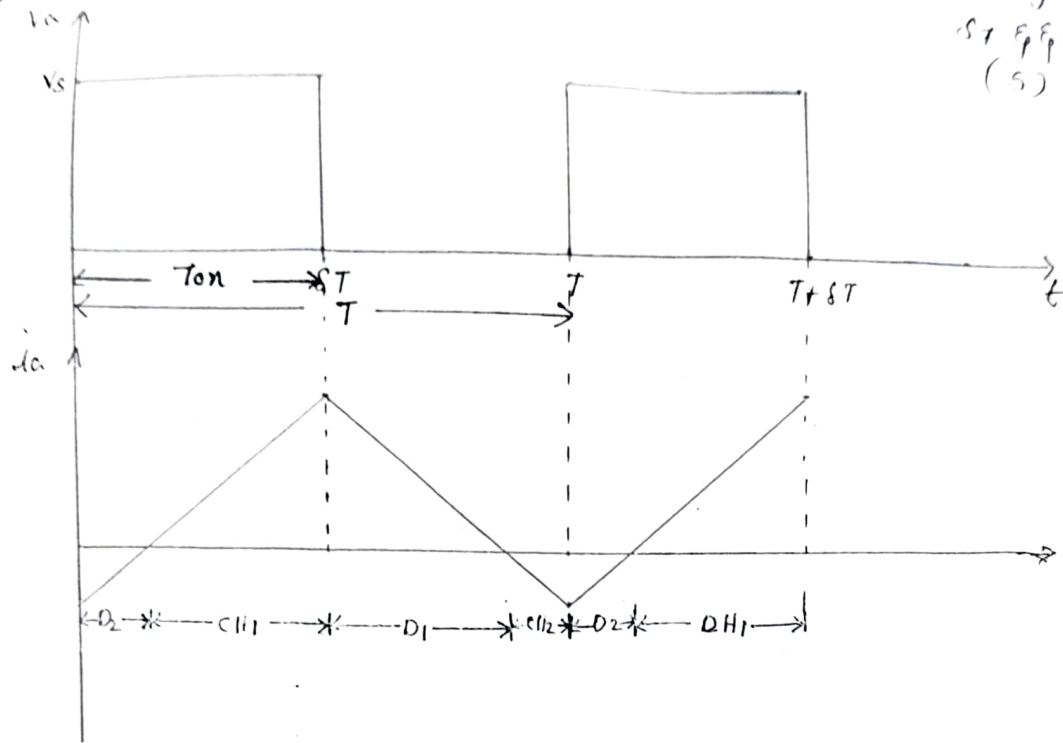
T

Two-quadrant chopper drive for separately excited.

dc motor

motoring & regenerative braking control.





During forward motoring the power is controlled by the chopper, while armature voltage and current both positive and power flows from dc source to the armature of the separately excited dc motor. On the other hand, during the regenerative braking the armature voltage being remains positive and the direction of armature current is reversed. In two quadrant dc motor drive, both motoring and braking mode are carried by one chopper configuration, which consist of two choppers CH_1 , CH_2 and two diodes.

Motoring mode.

When chopper CH_1 is ON, the supply voltage V_s gets connected to the armature terminals and therefore armature current i_a increases. When CH_1 OFF i_a freewheels through D_1 and therefore i_a decays. With CH_1 and D_1 , motor control in first quadrant is obtained.

forward braking mode

when α is turned on, the motor acts as generator and the armature current rises. when α is turned off β gets turned on and there for current increases in the opposite direction.

The average armature voltage for forward motoring operation

$$V_a = \cancel{E} V_s \alpha$$

while writing motor equation

$$\cancel{E} V_s \alpha = E + I_a R_a$$

$$\text{So armature current } I_a = \frac{V_s \cancel{E} \alpha - E}{R_a}$$

for motoring operation I_a is +ve.

$$\text{So } V_s \alpha > E$$

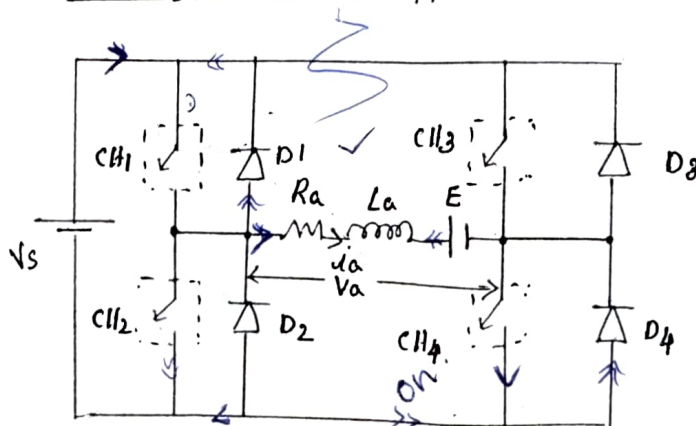
$$\alpha > \frac{E}{V_s}$$

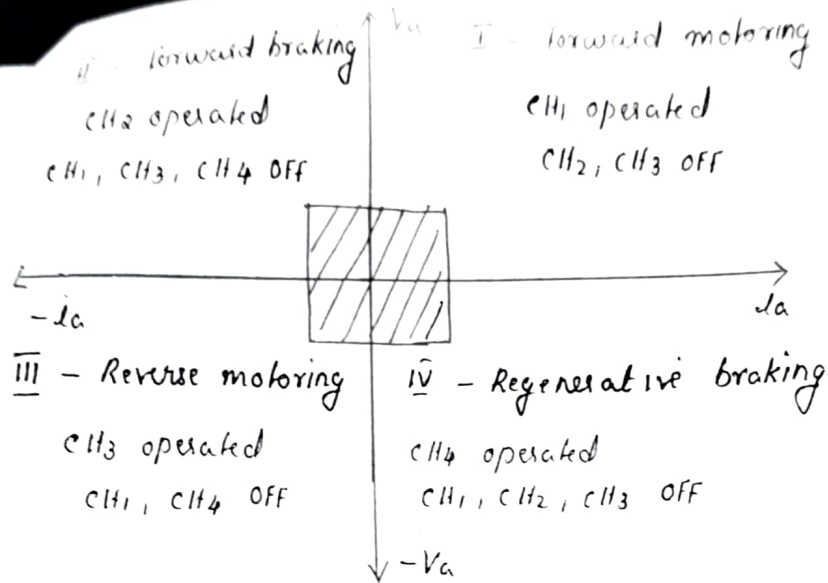
In braking operation armature current I_a is

$$\text{negative so } V_s \alpha < E$$

$$\alpha < \frac{E}{V_s}$$

Four quadrant chopper drive





In four quadrant chopper drive, motor can made to work in forward motoring (Ist quadrant), forward braking (IInd quadrant), reverse motoring (IIIrd quadrant) and reverse braking mode (IVth quadrant). The drive consist of four choppers, four diodes and separately excited dc motor.

(Ist quadrant - Forward motoring operation)

During forward motoring mode, choppers CH_2 and CH_3 are kept off, CH_4 is kept on whereas CH_1 is operated. when choppers CH_1 and CH_4 are turned ON, current flows through the path $V_s^+ - CH_1 - \text{load} - CH_4 - V_s^-$. Since both V_o & I_o are positive, we get the first quadrant operation. $\therefore$ when CH_1 turned OFF the current will freewheels through $D_2 - R_{\text{dca}} - E - CH_4$.

~~CH_1 is turned OFF the current will freewheel through $D_2 - R_{\text{dca}} - E - CH_4$.~~

IInd quadrant - Forward braking mode.

In this mode CH_1 , CH_3 and CH_4 are kept off and CH_2 is operated. when CH_2 is turned ON negative armature current rises through

$$-E - R_a - L_a - CH_2 - D_4$$

when CH_2 OFF the diodes D_1 & D_4 are turned on and the current will flow through

$$(-V_s) - D_4 - E - L_a - R_a - D_1 - (+ve)$$

and the motor acting as generator returns energy to the dc source. This results in forward braking mode in the II^{nd} quadrant.

III^{rd} quadrant - Reverse motoring mode

This operating mode is opposite to forward motoring mode. Choppers CH_1 , CH_4 are kept OFF. CH_2 is kept ON where as CH_3 is operated. When choppers CH_2 and CH_3 are turned ON, current flows through the path $V_s^+ - CH_3 - E - R_a - L_a - CH_2 - V_s^-$. Since both V_a and I_a are negative, we get third quadrant operation. When CH_3 is turned OFF, the negative armature current free wheels through

$$-E - R_a - L_a - CH_2 - D_4$$

IV^{th} quadrant - Reverse braking mode.

For this operating mode, CH_1 , CH_2 and CH_3 are kept OFF where as CH_4 is operated. When CH_4 is ON positive armature current rises through

$$-R_a - L_a - E - D_3 CH_4 - D_2$$

When CH_4 turned OFF, diodes D_3 and D_2 will conduct and the motor will act as a generator returns energy to the dc source.

$$(-V_s) - D_2 - R_a - L_a - E - D_3 - V_s$$

This leads to the regenerative braking operation of separately excited dc motor.